

ECE 514 Data Mining Project Proposal Fall-2026

Team: Data Sailors

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Dry Bean Classification

1. Question:

Can a bean producer use the physical measurements of a dry bean to identify its variety and separate mixed beans into the correct groups? The decision maker would be the bean producer or processing operator. The decision is which variety group each bean should be placed into. This can help separate mixed beans and produce more uniform groups.

2. Task:

This is a **classification** problem because each bean needs to be classified as one of seven known varieties: Seker, Barbunya, Bombay, Cali, Dermason, Horoz, or Sira. Classification is a good choice because the output is a category. Regression would not fit because we are not predicting a numerical value. Clustering is also not needed because the bean varieties are already known and labeled.

3. Label:

The target column is **Class**, which gives the variety of each bean. The researchers collected seven registered varieties of dry beans and used images of the beans to obtain their physical measurements. The purpose was to distinguish between bean varieties that can look similar. The available documentation does not clearly explain who assigned the Class label to each individual bean or exactly when it was assigned. The Class column is intended to directly represent the bean variety, so it is not being used as a proxy for something else.

4. Decision Threshold:

We will start with **80% predicted probability** as the decision threshold. If the model is at least 80% confident that a bean belongs to one variety, it can be placed into that variety group. If the highest probability is below 80%, the bean would be sent for additional inspection instead of being automatically classified. The 80% value is a starting point and can be evaluated later.

5. Data Provenance:

The **Dry Bean Dataset** comes from the UCI Machine Learning Repository. It contains **13,611 beans with 16 numerical features** based on the size and shape of each bean. There are seven bean classes and no reported missing values. This is more than a small classroom dataset because it contains over 13,000 real samples, several physical measurements, and seven similar bean varieties.

6. Feared Failure Mode:

The failure mode we are most concerned about is the **deployment gap**. The measurements in the dataset were created from bean images using a specific computer-vision setup. If a different camera, lighting condition, or image-processing method is used in the real world, the measurements could be different and the model may not work as well.

Data Set Link:

<https://archive.ics.uci.edu/dataset/602/dry+bean+dataset>

Published Article Link:

<https://www.sciencedirect.com/science/article/pii/S0168169919311573>